

Network Traffic Simulation Tool - Business Plan

Converted from the uploaded Business Plan source document

Executive Summary

An interactive web-based network simulation tool designed to help students, network administrators, and IT professionals visualize and understand real network behavior including routing protocols, packet queueing, bandwidth constraints, and Quality of Service (QoS). The tool balances accessibility with technical accuracy, making it suitable for both educational and professional training contexts.

Product Overview

What It Does

An interactive canvas-based network simulator where users create network topologies and observe realistic packet behavior in real-time. Unlike toy simulators, this tool models actual networking concepts while remaining accessible to non-experts.

Key Features

- **Build Network Topologies:** Drag-and-drop routers, switches, and hosts
 - **Configure Links:** Set bandwidth (Mbps), latency (ms), and packet loss rates
 - **Realistic Routing:** Implements actual routing algorithms (OSPF, static routes)
 - **QoS Simulation:** Priority queueing for different traffic types (voice, video, data)
 - **Live Metrics:** Throughput, latency, jitter, packet loss visualization
 - **Traffic Generators:** HTTP, VoIP, video streaming, bulk data transfer patterns
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Technical Specifications

Network Elements

Nodes

Routers

- Forward packets based on routing tables
- Support static routes or dynamic routing (OSPF-like)
- Have configurable buffer sizes
- Implement weighted fair queueing or priority queueing

Switches

- Layer 2 forwarding (MAC address tables)
- VLAN support (optional advanced feature)
- No routing decisions

Hosts

- Traffic sources and destinations
- Generate realistic traffic patterns

Links

Properties

- **Bandwidth:** 1 Mbps to 10 Gbps (dropdown: 1M, 10M, 100M, 1G, 10G)
- **Latency:** 1-500ms (propagation delay)
- **Packet Loss:** 0-5% random loss rate
- **Duplex:** Full-duplex (bidirectional)

Traffic Types

Users can generate different traffic classes:

1. **Voice (VoIP):**
 - Small packets (~200 bytes), constant bitrate (~64 Kbps)
 - High priority, latency-sensitive (<150ms requirement)
2. **Video Streaming:**
 - Variable packet sizes, moderate bitrate (1-5 Mbps)
 - Medium priority, sensitive to jitter
3. **Web/HTTP:**
 - Bursty traffic pattern
 - Medium priority, tolerant of some delay
4. **Bulk Data:**
 - Large packets (1500 bytes), high throughput
 - Low priority, delay-tolerant

Routing

Static Routing

- Users manually configure routes
- Shows routing table for each router
- Simple to understand, full control

Dynamic Routing (OSPF-like)

- Routers automatically discover shortest path
- Cost metric based on bandwidth
- Adapts to link failures
- Educational visualization of route convergence

Quality of Service (QoS)

Priority Queuing

- Each router has multiple queues (High, Medium, Low)
- Traffic types map to priority levels
- High-priority packets served first
- Shows how VoIP stays smooth even during congestion

Visualization

- Color-coded packets by type (Voice=red, Video=blue, Web=green, Bulk=gray)
 - Queue depth indicators on routers
 - Per-class throughput graphs
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Simulation Mechanics

Packet Flow (200ms tick)

5. **Generation:** Traffic generators create packets at realistic rates
6. **Routing Decision:** Router looks up destination in routing table
7. **Queuing:** Packet enters appropriate priority queue
8. **Transmission:**
 - Packet transmission time = $(\text{packet_size_bits} / \text{link_bandwidth_bps})$
 - Only one packet transmits per link at a time
9. **Propagation:** Packet travels with link latency delay
10. **Reception:** Arrives at next hop, process repeats
11. **Metrics:** Track per-packet latency, calculate aggregate throughput

Congestion Behavior

Queue Management

- Each router has finite buffer (e.g., 50 packets)
- When buffer full: tail drop (reject new packets)
- Shows dropped packet count per router

TCP-like Behavior (optional)

- Bulk data sources slow down when detecting loss
- VoIP sources don't adjust (constant rate)
- Demonstrates why congestion affects traffic differently

Real-World Accuracy

What's Realistic

- [Yes] Packet transmission times based on actual bandwidth
- [Yes] Propagation delay modeled
- [Yes] Priority queuing matches real QoS
- [Yes] Routing algorithms follow OSPF principles

- [Yes] Traffic patterns match real applications

Simplifications (for usability)

- [Note] No TCP windowing or flow control complexity
 - [Note] Simplified packet headers (no full IP/Ethernet simulation)
 - [Note] Deterministic rather than stochastic in some areas
 - [Note] No physical layer details (fiber vs copper)
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User Experience Flow

Scenario 1: Student Learning Routing

12. Create 4 routers in a mesh topology
13. Configure static routes manually
14. Generate traffic between hosts
15. **Observe:** Packets follow configured paths
16. Break a link (simulate failure)
17. **See:** Traffic fails, routing table becomes incomplete
18. Switch to OSPF mode
19. **Watch:** Routes automatically reconverge around failure

Learning outcome: Understanding routing protocols vs static routes

Scenario 2: Network Admin QoS Planning

20. Build office network (1 router, 3 switches, multiple hosts)
21. Simulate normal business traffic (web, email)
22. Add VoIP phones
23. **Problem:** Without QoS, voice quality degrades during file transfers
24. Enable priority queuing
25. **Solution:** Voice packets get priority, quality improves
26. Export metrics report showing latency by traffic class

Learning outcome: Justifying QoS implementation to management

Scenario 3: Capacity Planning

27. Load existing network topology
28. Generate baseline traffic
29. Gradually increase video conferencing load
30. **Observe:** Link utilization graphs, identify bottleneck
31. Upgrade bottleneck link bandwidth
32. **Validate:** Congestion resolved, metrics improve
33. Export "before/after" comparison

Learning outcome: Data-driven network upgrade decisions

Target Market

Primary Users

1. Educational Institutions

- Computer networking courses (undergraduate/graduate)
- Cisco CCNA/CCNP exam preparation
- High school STEM programs
- **Market size:** 4,000+ universities, 20,000+ high schools in US

2. Corporate Training

- IT department onboarding
- Network certification bootcamps
- Non-technical staff education (help desk, sales engineers)
- **Market size:** Fortune 500 companies, MSPs, tech consultancies

3. Individual Learners

- Self-taught network engineers
- Certification candidates
- Hobbyists/homelab enthusiasts

Why They Need This

Problems it solves

- Physical lab equipment is expensive (\$10K+ for Cisco lab)
- Existing simulators (Packet Tracer) are complex and slow
- Abstract concepts (QoS, routing) are hard to visualize
- Need quick "what-if" analysis without real hardware

Competitive advantages

- **Web-based:** No installation, works on Chromebooks
 - **Visual:** Real-time animation shows packet flow
 - **Fast:** Iterate on designs in seconds
 - **Accurate enough:** Real networking concepts, not oversimplified
 - **Affordable:** Freemium model vs expensive software licenses
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Business Model

Pricing Strategy

Free Tier

- Up to 10 nodes
- Basic traffic types (web, bulk data)
- Static routing only
- Community support
- **Goal:** User acquisition, educational access

Pro Tier (\$9/month or \$90/year)

- Unlimited nodes
- All traffic types (VoIP, video)
- QoS and advanced routing
- Save/export topologies
- Metric reports (PDF export)
- Email support
- **Target:** Individual professionals, students

Team/Education Tier (\$299/year per classroom)

- All Pro features
- Classroom management (teacher dashboard)
- Pre-built lab scenarios
- Student progress tracking
- Priority support
- **Target:** Schools, training programs

Enterprise Tier (Custom pricing)

- Self-hosted option
- White-labeling
- Custom integrations
- Dedicated support
- Training workshops
- **Target:** Large corporations, universities

Revenue Projections (Year 1)

Assumptions

- 10,000 free users
- 2% convert to Pro (\$9/mo) = 200 users x \$108/yr = \$21,600
- 20 educational institutions x \$299/yr = \$5,980
- 2 enterprise deals x \$5,000/yr = \$10,000

Total Year 1 Revenue: ~\$37,500

Year 3 Target: 50,000 users, 1,000 Pro (\$108K), 100 institutions (\$30K), 10 enterprise (\$50K) = **\$188K**

Development Roadmap

MVP (Months 1-3)

- Basic canvas with routers and hosts
- Link configuration (bandwidth, latency)
- Static routing
- Single traffic type (bulk data)
- Throughput visualization
- **Goal:** Validate core concept

Version 1.0 (Months 4-6)

- Multiple traffic types
- Priority queuing
- OSPF-like routing
- Packet loss simulation
- Metrics dashboard
- Save/load topologies
- **Goal:** Launch freemium model

Version 2.0 (Months 7-12)

- Pre-built scenarios library
- PDF metric reports
- Team features (sharing)
- Advanced queuing (WFQ)
- Tutorial system
- **Goal:** Scale to 1,000+ users

Future Features (Year 2+)

- Mobile app version
 - Real-time collaboration
 - AI-powered network design suggestions
 - Integration with real network monitoring tools
 - Physical device emulation mode
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Technical Architecture

Frontend

- **React + TypeScript** for UI
- **Canvas API** or **SVG** for visualization
- **D3.js** for graphs and metrics
- Runs entirely in browser (no server required for simulation)

Backend (Optional)

- **Node.js + Express** for user accounts
- **PostgreSQL** for saving topologies
- **AWS S3** for file storage
- **Stripe** for payments

Simulation Engine

- Written in TypeScript
 - Event-driven architecture (discrete event simulation)
 - ~60 FPS rendering for smooth animation
 - Optimized for 100+ nodes without lag
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Competitive Analysis

Feature	Our Tool	Cisco Packet Tracer	GNS3	NS-3
Web-based	✓	✗ (desktop)	✗ (desktop)	✗ (command-line)
Easy setup	✓	⚠ (moderate)	✗ (complex)	✗ (very complex)
Visual packets	✓	⚠ (limited)	✗	✗
QoS simulation	✓	✓	✓	✓
Speed	✓ Fast	⚠ Slow	⚠ Slow	✓ Fast
Cost	Freemium	Free (education)	Free	Free
Accuracy	⚠ Good enough	✓ High	✓ Very high	✓ Research-grade
Target user	Students + pros	Cisco students	Network engineers	Researchers

Our niche: Balance of accuracy and usability. Easier than Packet Tracer, more realistic than toy simulators.

Success Metrics

Year 1 KPIs

- 10,000 registered users
- 200 paying subscribers
- 20 educational institution customers
- 70% user retention (90 days)
- Average session time: 15+ minutes

User Engagement

- Topologies created per user
- Scenarios completed
- Feature adoption rates (QoS, routing)
- Support ticket volume

Financial

- Monthly Recurring Revenue (MRR)
 - Customer Acquisition Cost (CAC)
 - Lifetime Value (LTV)
 - Churn rate
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Risk Analysis

Technical Risks

- **Browser performance:** Mitigation: Optimize for 100 nodes, warn users beyond that
- **Simulation accuracy:** Mitigation: Clear documentation of simplifications
- **Cross-browser compatibility:** Mitigation: Test on Chrome, Firefox, Safari, Edge

Market Risks

- **Low adoption:** Mitigation: Freemium model reduces barrier
- **Cisco competition:** Mitigation: Focus on ease-of-use and web-based advantage

- **Educational budget cuts:** Mitigation: Individual pricing tier as backup

Business Risks

- **Feature creep:** Mitigation: Stick to roadmap, prioritize MVP
 - **Support burden:** Mitigation: Comprehensive docs, community forum
 - **Scaling costs:** Mitigation: Client-side simulation minimizes server load
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Team & Resources Needed

Founding Team

- **Technical Lead:** Full-stack dev (React + backend)
- **Network SME:** Design realistic simulation algorithms
- **Designer/UX:** Create intuitive interface

Year 1 Budget

- Development (3 months x \$10K/mo): \$30,000
- Infrastructure (AWS, domain, tools): \$2,000
- Marketing (content, ads): \$5,000
- Legal (incorporation, terms): \$2,000
- **Total:** ~\$40,000

Funding Strategy

- Bootstrap with founders' time (nights/weekends)
 - Launch on Product Hunt for initial users
 - Apply for education-focused grants
 - Seed funding if traction is strong
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Conclusion

This network simulation tool fills a gap between oversimplified toys and complex professional tools. By modeling real networking concepts (bandwidth, latency, QoS, routing) in an accessible visual interface, it serves students learning fundamentals and professionals planning networks.

The web-based approach removes installation friction, the freemium model reduces financial barriers, and the focus on visualization makes abstract concepts concrete. With realistic technical foundations and clear educational value, this tool can become the go-to platform for interactive network learning.

Next Steps

34. Build MVP with static routing and basic traffic
35. Beta test with local university networking class
36. Iterate based on feedback
37. Launch freemium version
38. Scale marketing to reach 10K users in Year 1